

Functional Central Limit Theorems for Local Statistics of Spatial Birth-Death Processes in the Thermodynamic Regime

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(Joint work w. Omer Bobrowski and Robert J. Adler)
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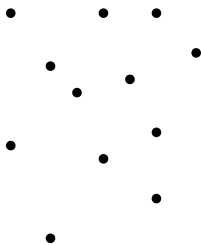
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Gniezno

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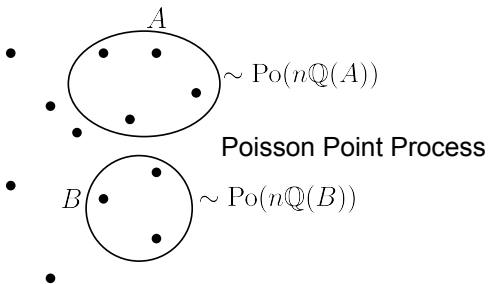


Poisson Point Process

$$t = 0$$

Spatial Birth Death Processes

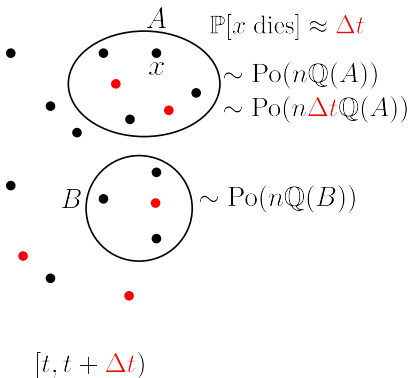
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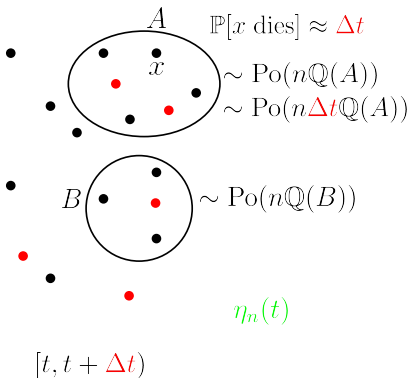
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Functionals of Interest - I

$$f_n(t) := \sum_{\mathcal{S} \subset \eta_n(t)} \xi_{k,r}(\mathcal{S})$$

for $\xi_{k,r} : \text{Fin}(\mathbb{R}^d) \rightarrow [0, \infty)$ s.t.

- **Fixed Size:** $\xi_{k,r}(\mathcal{S}) = 0$ if $|\mathcal{S}| \neq k$
- **Local:** $\exists \rho > 0$ s.t. $\text{diam}(\mathcal{S}) > \rho r$ implies $\xi_{k,r}(\mathcal{S}) = 0$
- **Trans. and Scale Inv.:** $\xi_{k,r}(\alpha \mathcal{S} + x) = \xi_{k,r/\alpha}(\mathcal{S})$
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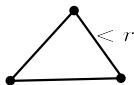
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$$\xi_{k,r}(\mathcal{S}) = 1\{\mathcal{S} \text{ is isomorphic to } \mathcal{G}\}$$

Functionals of Interest - II

$$F_n(t) := \sum_{\mathcal{S} \subset \eta_n(t)} \xi_{k,r}(\mathcal{S}) \mathbf{1} \{ (\eta_n(t) \setminus \mathcal{S}) \cap N_r(\mathcal{S}) = \emptyset \}$$

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- ▶ **Local:** $\exists \beta > 0$ s.t. for all $\mathcal{S} \in \text{Fin}(\mathbb{R}^d)$

$$\text{diam}(N_r(\mathcal{S})) \leq \beta \max(r, \text{diam}(\mathcal{S}))$$

- ▶ **Feasibility:** $\exists$ a nonempty open set $\mathcal{B} \subset (\mathbb{R}^d)^{k-1}$ s.t.

$$\xi_{k,1}(0, y_1, \dots, y_{k-1}) \text{vol}[N_1(0, y_1, \dots, y_{k-1})] > 0$$

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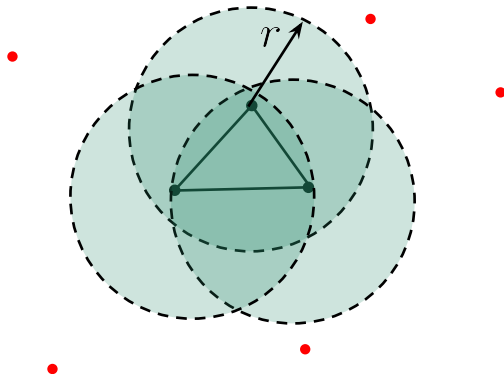
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for some $c_j > 0$. Furthermore, the limiting process has a.s. continuous paths.

Proof Idea

1. Convergence of finite dimensional distributions

2. Tightness

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► Cramer-Wold

$$\sum_{i=1}^m \omega_i f_n(t_i) = \sum_{\mathcal{S} \subset \eta(n,T)} \xi_{k,r}(\mathcal{S}) \sum_{i=1}^m \omega_i \mathbb{1}\{\text{Each point in } \mathcal{S} \text{ alive at } t_i\}$$

for all $0 \leq t_1 \leq \dots \leq t_m \leq T$.

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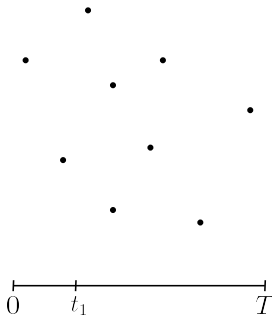
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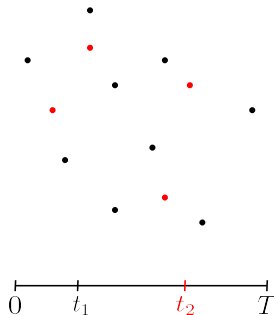
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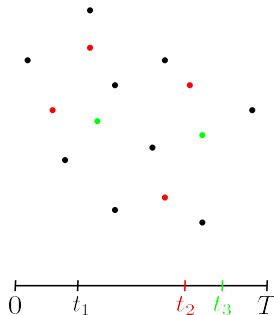
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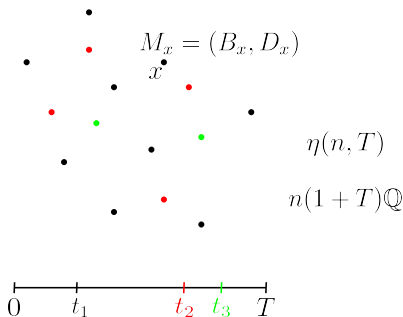
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► 4th moment bounds.

Normal Approximation for Stabilizing Functionals (Lachièze-Rey, Schulte, and Yukich 2019)

$$H_n := \sum_{x \in \eta_n} h_n(x, \eta_n)$$

Corollary 2.2

Assume that $(h_n)_{n \geq 1}$ are stabilized with a decaying radius, and have bounded $(4 + \varepsilon)$ th moment. Assume also $\sup_{n \geq 1} \frac{n}{\text{var } H_n} \leq C$ for some $C > 0$. Then

$$d_{\text{Kol}} \left(\frac{H_n - \mathbb{E} H_n}{\sqrt{\text{var } H_n}}, Z \right) \leq \frac{C'}{\sqrt{\text{var } H_n}} \quad \text{for } n \geq 1,$$

for some $C' > 0$ where $Z \sim N(0, 1)$.

Moment Calculations - Definitions

Definition (Intersection Pattern)

Let $\mathcal{I}_\ell$ denote a collection of natural numbers indexed by nonempty subsets of $[\ell] = \{1, \dots, \ell\}$, i.e. $\mathcal{I}_\ell = (I_J)_{J \subset [\ell], J \neq \emptyset}$. Given a sequence of sets $\mathcal{S}_1, \dots, \mathcal{S}_\ell \in \text{Fin}(\mathbb{R}^d)$, suppose that for all nonempty $J \subset [\ell]$ we have

$$\left| \left(\bigcap_{j \in J} \mathcal{S}_j \right) \cap \left(\bigcap_{j \notin J} (\mathcal{S}_j)^c \right) \right| = I_J,$$

In this case we say that $\mathcal{S}_1, \dots, \mathcal{S}_\ell$ obey the **intersection pattern** $\mathcal{I}_\ell$, and denote this by $(\mathcal{S}_1, \dots, \mathcal{S}_\ell) \in \mathcal{I}_\ell$. Set

$$|\mathcal{I}| := \sum_J I_J = |\mathcal{S}_1 \cup \dots \cup \mathcal{S}_\ell|.$$

Moment Calculations - Definitions

Definition

Given a sequence of points $\mathbf{x} = (x_1, \dots, x_{|\mathcal{I}|})$ in $\mathbb{R}^d$, let

$$\Pi_{\mathcal{I}}(\mathbf{x}) := (\mathcal{T}_1, \dots, \mathcal{T}_{\ell})$$

be a splitting of the tuple $\mathbf{x}$ into $(\mathcal{T}_1, \dots, \mathcal{T}_{\ell}) \in \mathcal{I}$, in an arbitrary but fixed manner.

Moment Calculations - Key Lemmas

Lemma (Generalized Mecke Equation)

$$\mathbb{E} \left[\sum_{\mathcal{S}_1 \subset \eta_n} \cdots \sum_{\mathcal{S}_\ell \subset \eta_n} h(\bar{\mathcal{S}}, \eta_n) \mathbb{1}\{\bar{\mathcal{S}} \in \mathcal{I}\} \right] \asymp n^{|\mathcal{I}|} \mathbb{E} [h(\Pi_{\mathcal{I}}(\mathbf{X}), \mathbf{X} \cup \eta'_n)]$$

where $\bar{\mathcal{S}} := (\mathcal{S}_1, \dots, \mathcal{S}_\ell)$, and $\mathbf{X}$ is a tuple of $|\mathcal{I}|$ iid points in $\mathbb{R}^d$, with density q , and independent of η'_n .

Moment Calculations - Definitions

Definition

Let $\#\mathcal{I}$ denote the **number of connected components** in the intersection graph of $\mathcal{I}$, i.e., the graph with ℓ nodes, where each node pair (i, j) is connected by an edge if and only if $|\mathcal{T}_i \cap \mathcal{T}_j| > 0$.

Moment Calculations - Key Lemmas

Lemma (Moments of local functions)

Let $\mathcal{I} = \mathcal{I}_\ell$ be an intersection pattern, such that if $(\mathcal{S}_1, \dots, \mathcal{S}_\ell) \in \mathcal{I}_\ell$ then $|\mathcal{S}_j| = k_j \geq 2$ for every $1 \leq j \leq \ell$. Let $\mathbf{X}$ be a sequence of $|\mathcal{I}|$ iid points in $\mathbb{R}^d$ with density q , and let $(\mathcal{T}_1, \dots, \mathcal{T}_\ell) = \Pi_{\mathcal{I}}(\mathbf{X})$. Then

$$\mathbb{E} \left[\prod_{j=1}^{\ell} \xi_{k_j, r}(\mathcal{T}_j) \right] \asymp r^{d(|\mathcal{I}| - \#\mathcal{I})}.$$

Tightness

By Theorem 13.5 of *Convergence of Probability Measures* (Billingsley 1999), sufficient condition for tightness of $X_n(t)$:

$$\mathbb{E} [(X_n(t) - X_n(s))^2 (X_n(s) - X_n(r))^2] \leq C(t - r)^2$$

for all $0 \leq r \leq s \leq t$ and a constant $C > 0$.

Thank you.

Spatial Birth Death Processes

- ▶ Chris Preston (1975). “Spatial birth-and-death processes”. *Bull. Inst. Internat. Statist.*
- ▶ Lorenzo Bertini, Nicoletta Cancrini, and Filippo Cesi (2002). “The spectral gap for a Glauber-type dynamics in a continuous gas”. *Ann. Inst. H. Poincaré Probab. Statist.*
- ▶ M. N. M. van Lieshout (1994). “Stochastic annealing for nearest-neighbour point processes with application to object recognition”. *Adv. in Appl. Probab.*
- ▶ A. J. Baddeley and M. N. M. van Lieshout (1995). “Area-interaction point processes”. *Ann. Inst. Statist. Math.*

Malliavin-Stein bounds

- ▶ Ivan Nourdin and Giovanni Peccati (2009). “Stein’s method on Wiener chaos”. *Probab. Theory Related Fields*
- ▶ G. Peccati, J. L. Solé, M. S. Taqqu, and F. Utzet (2010). “Stein’s method and normal approximation of Poisson functionals”. *The Annals of Probability*
- ▶ Günter Last and Mathew D. Penrose (2011). “Poisson process Fock space representation, chaos expansion and covariance inequalities”. *Probab. Theory Related Fields*

Normal Approximation for Stabilizing Functionals (Lachièze-Rey, Schulte, and Yukich 2019)

Definition (Decaying stabilization)

Let $R_n \geq 0$ with $\limsup_{n \rightarrow \infty} n(R_n)^d < \infty$. We say that R_n is a radius of decaying stabilization for the (local) function $h_n : \mathbb{R}^d \times \text{Fin}(\mathbb{R}^d) \rightarrow \mathbb{R}$, if, for all $x \in \mathbb{R}^d$ and $\mathcal{Y} \in \text{Fin}(\mathbb{R}^d)$,

$$h_n(x, (\mathcal{Y} \cup x) \cap B(x, R_n)) = h_n(x, \mathcal{Y} \cup x)$$

where $B(x, \lambda) := \{z \in \mathbb{R}^d : |z - x| \leq \lambda\}$ for $x \in \mathbb{R}^d$ and $\lambda > 0$.

Infinite superposition of OU processes can be discontinuous!

E. Csáki, M. Csörgő, Z. Y. Lin, and P. Révész (1991). “On infinite series of independent Ornstein-Uhlenbeck processes”. *Stochastic Process. Appl.*

Continuity of $\bar{F}_n(t)$

Continuity criterion:

- ▶ A stationary Gaussian process $X(t)$ is a.s. path-continuous if there exists a non-decreasing function $\psi(h)$ such that

$$\mathbb{E} [(X(h) - X(0))^2] \leq \psi^2(h) \quad \text{and} \quad \int_1^\infty \psi(e^{-x^2}) dx < \infty.$$

Due to

Xavier Fernique (1964). “Continuité des processus Gaussiens”. *C. R. Acad. Sci. Paris*